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(54) **DYNAMIC GENERATION OF DIGITAL
MESSAGES WITH UNIQUE LINKS FOR
DIRECT TO MERCHANT PAYMENTS**

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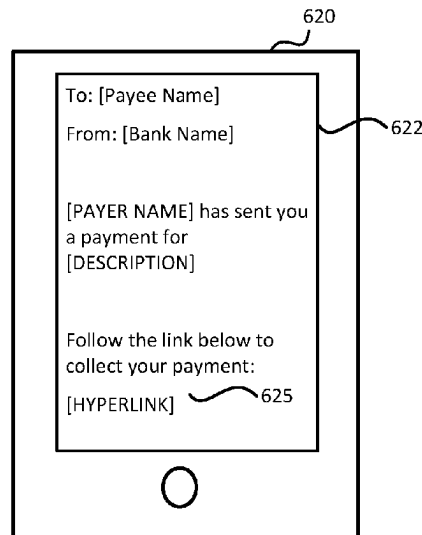
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(57) **ABSTRACT**

A payer initiates a payment transaction of a certain dollar amount through a user interface provided by their bank. The bank generates a hyperlink associated with the payment transaction. By managing a state of the transaction, the hyperlink can be given an expiration time and/or a counter of visits. The hyperlink is sent to a payment recipient in a text message or email, either by the payer or by the bank. Upon following the hyperlink, the recipient can enter their own debit card information to receive the money transfer to their debit account, without the need to set up an account login or account. A notification mechanism, such as a broadcast stream of payments, can be employed to inform different entities (such as those recording property payments or utilities payments) that a payment was made to the payment recipient.

20 Claims, 5 Drawing Sheets



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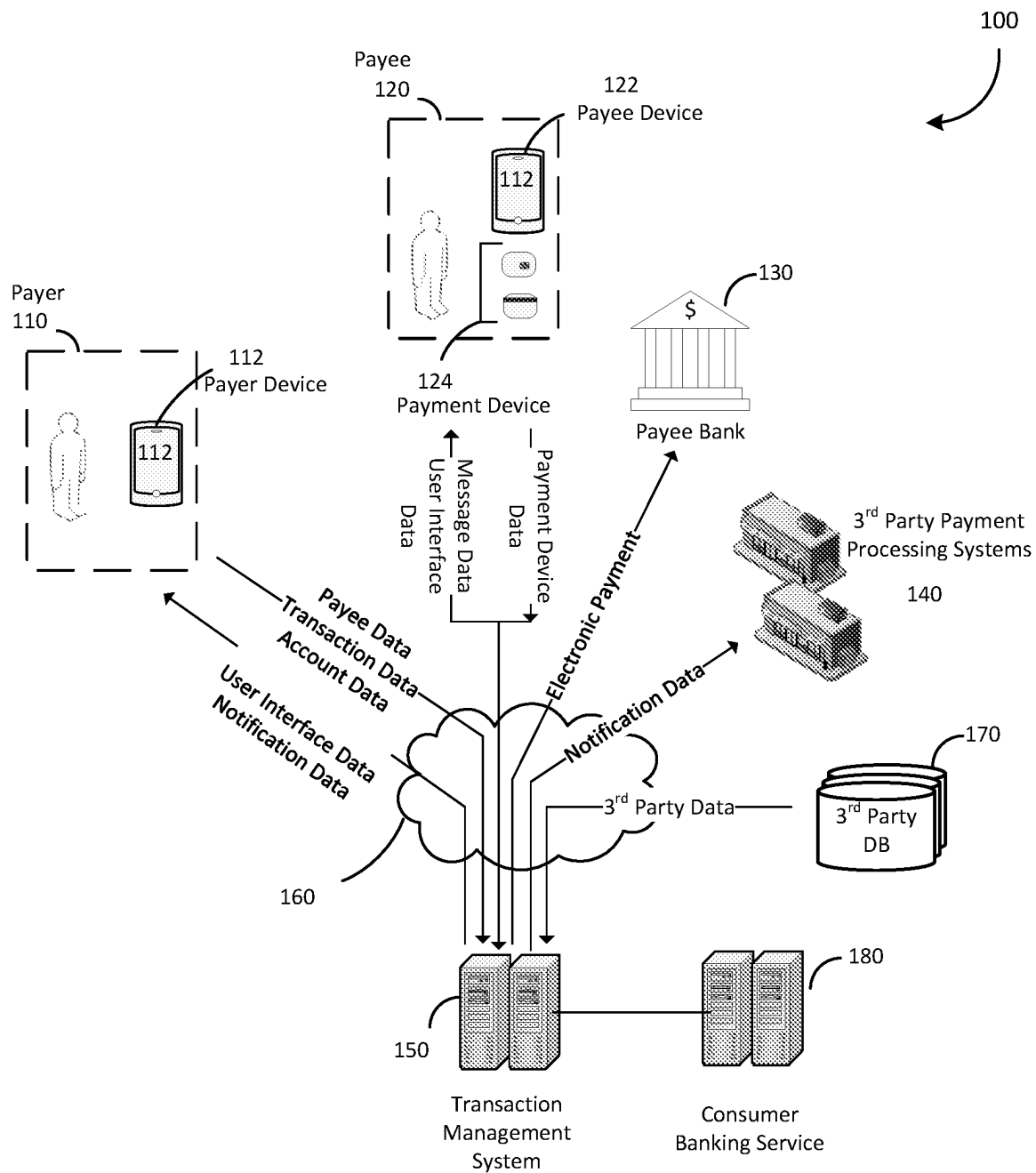


FIG. 1

112

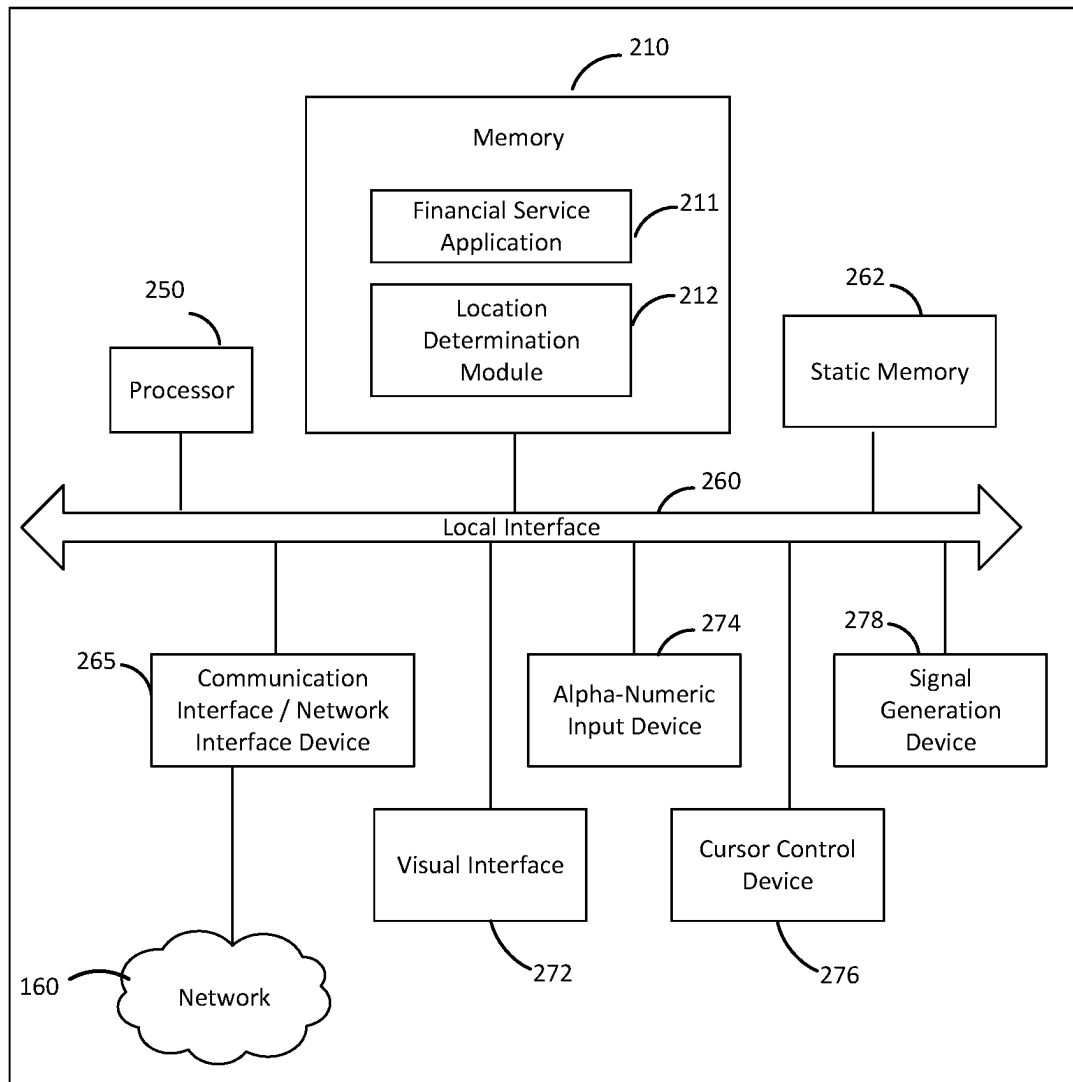
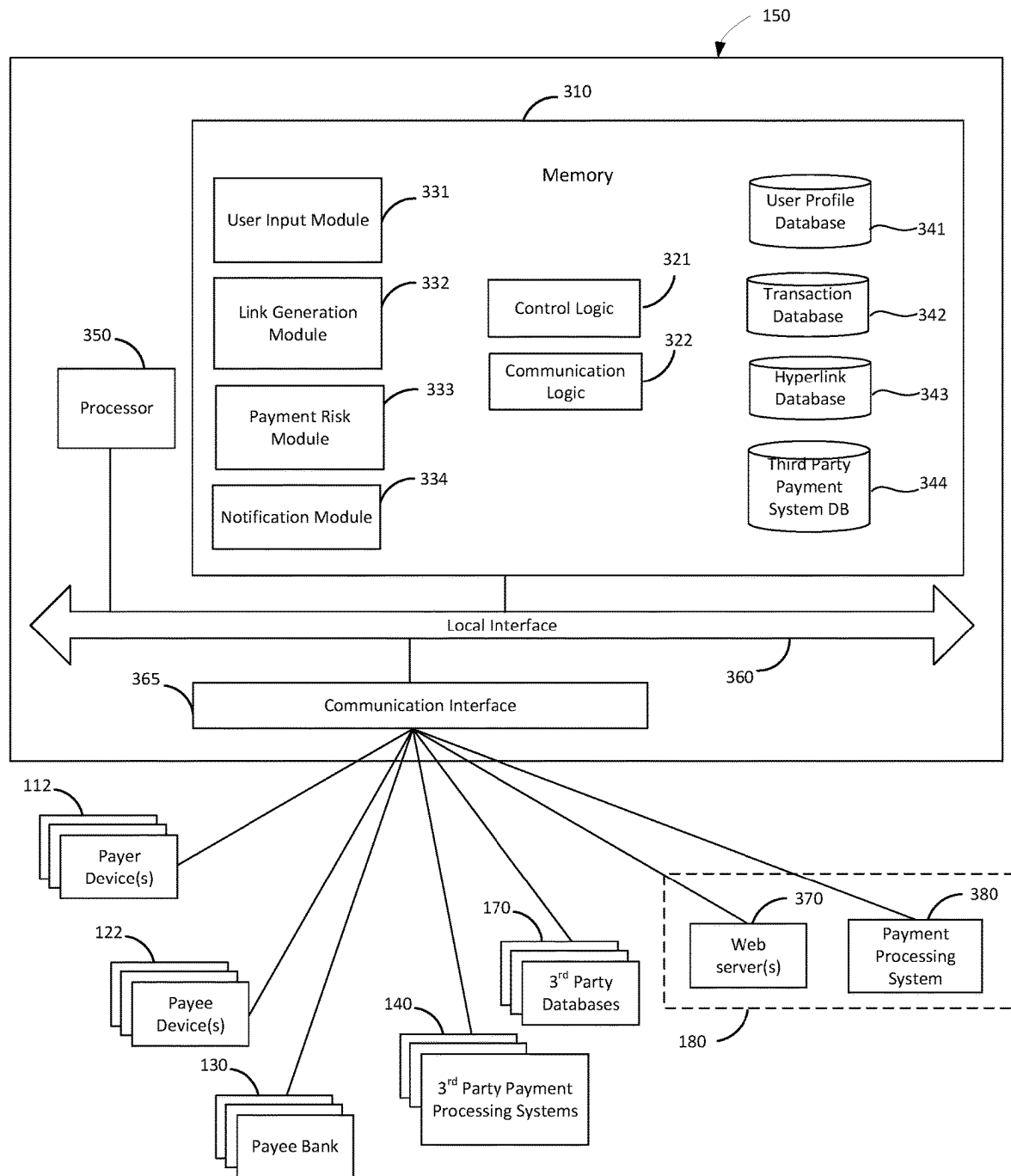


FIG. 2

**FIG. 3**

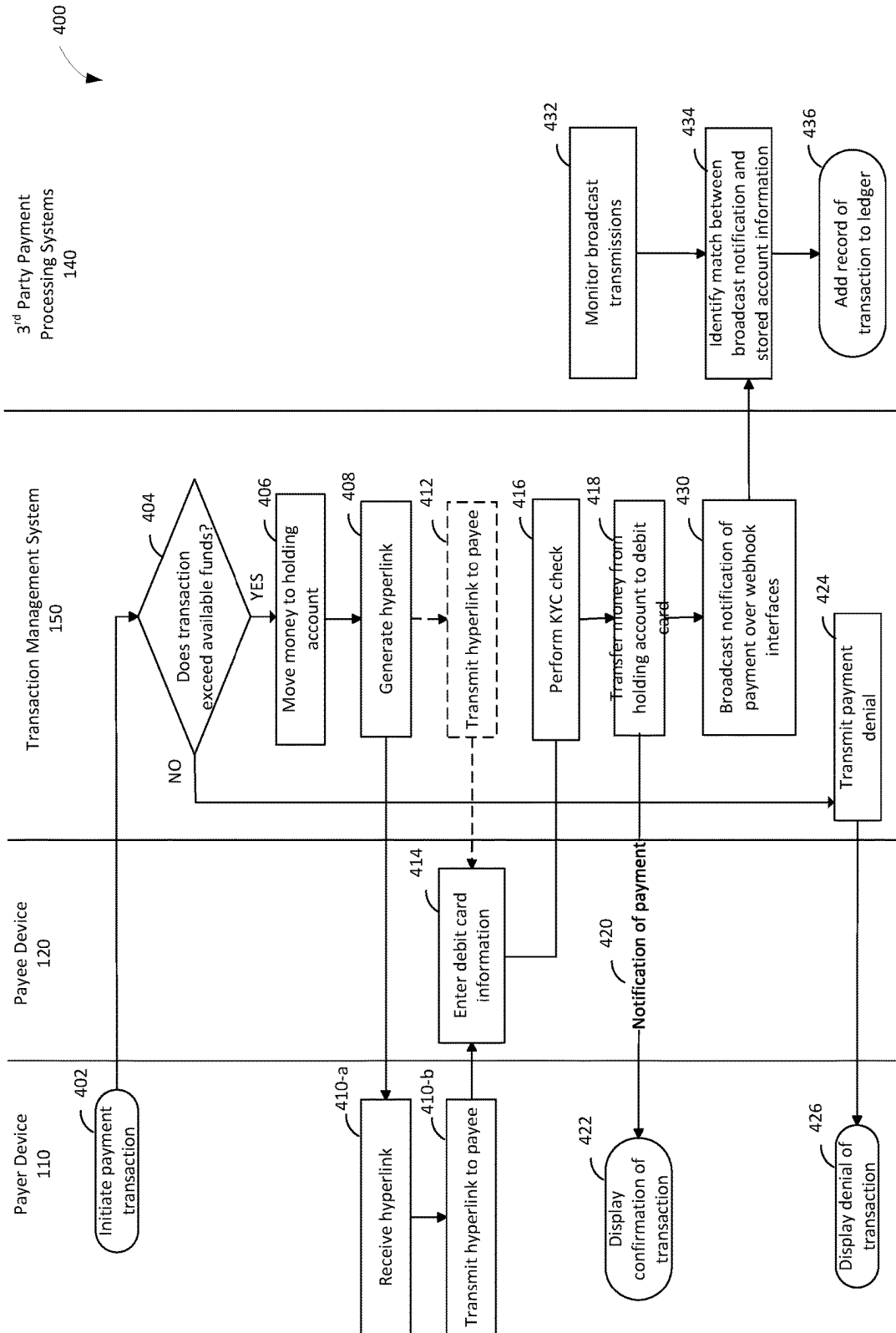
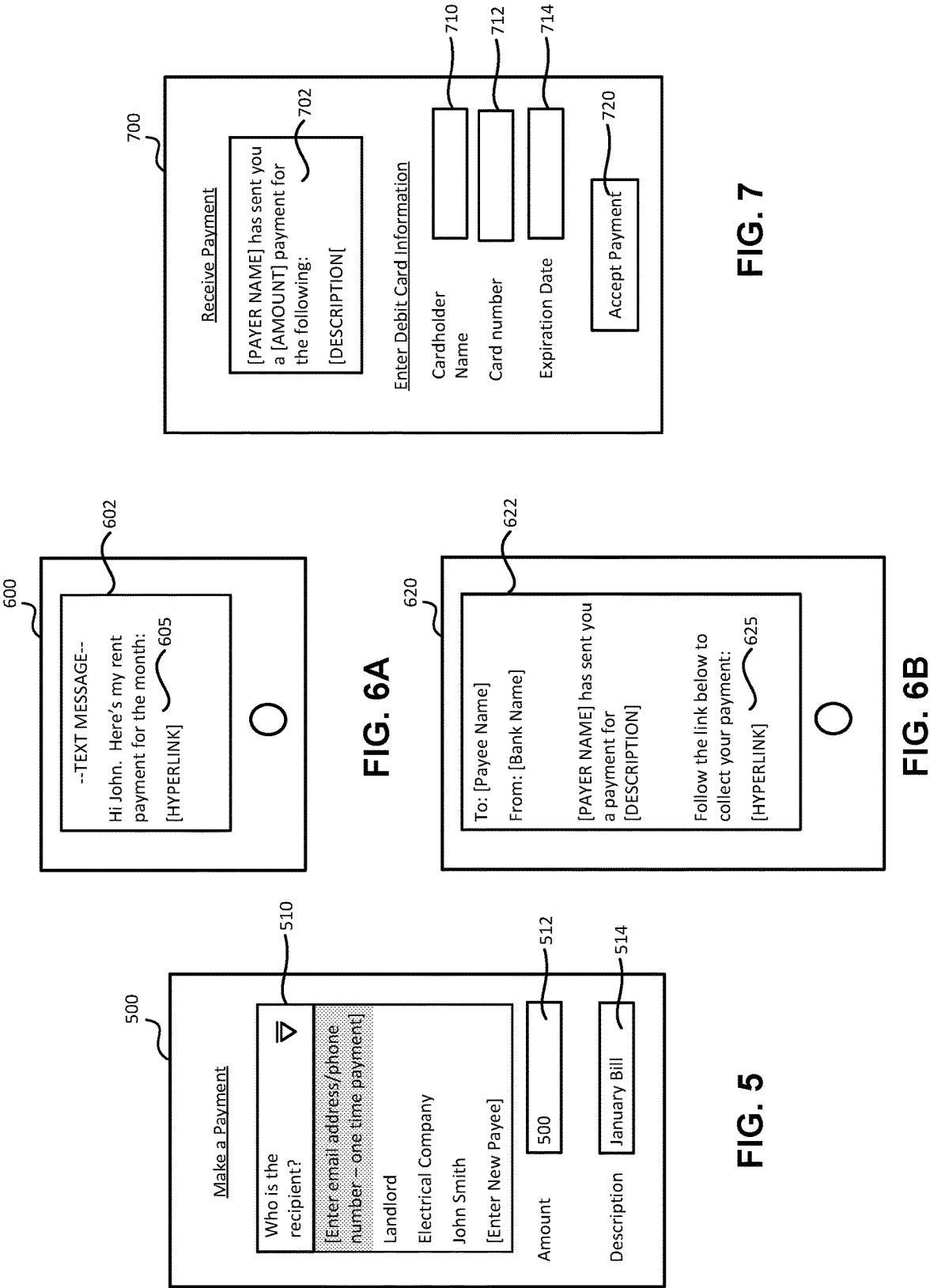


FIG. 4



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DYNAMIC GENERATION OF DIGITAL MESSAGES WITH UNIQUE LINKS FOR DIRECT TO MERCHANT PAYMENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/038,760, filed on Sep. 30, 2020. The aforementioned application is incorporated by reference in its entirety.

BACKGROUND

Consumers are increasingly offered an option for paperless billing and payments, that is, an option to pay bills through managed digital platforms. In a typical implementation, to facilitate the digital transfer of money a customer (payer) must create an account on a third party payment system and enter their payment information. The third party system acts as a middle man that facilitates a payment to a recipient (the intended payee). As one example, a tenant intending to pay their rent electronically must create an account with a property management company and pay the property management company via their third party software. The landlord, having their own account with the property management company, receives the money in due course. In another implementation, a payer uses their bank's website or application to instruct that their bank issue a check to the intended payee, who must then independently take that check and deposit it in a conventional manner.

Additional solutions that facilitate direct digital payments between source and target entities while maintaining security and convenience of transactions are generally desired.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosed embodiments have other advantages and features that will be more readily apparent from the detailed description, the appended claims, and the accompanying figures (or drawings). The disclosure can be better understood with reference to the following drawings. The elements of the drawings are not necessarily to scale relative to each other, emphasis instead being placed upon clearly illustrating the principles of the disclosure.

FIG. 1 is a diagram illustrating an environment including a transaction management system for facilitating payment transactions between a payer and a payee in accordance with some embodiments of the present disclosure.

FIG. 2 is a block diagram illustrating component parts of a payer device in accordance with some embodiments of the present disclosure.

FIG. 3 is a block diagram illustrating component parts of a transaction management system in accordance with some embodiments of the present disclosure.

FIG. 4 is a flow chart of an exemplary transaction management process in accordance with some embodiments of the present disclosure.

FIG. 5 illustrates an exemplary user interface for a user to input instructions to facilitate a payment, in accordance with some embodiments of the present disclosure.

FIGS. 6A and 6B illustrate exemplary messages transmitted to a payee containing a hyperlink, in accordance with some embodiments of the present disclosure.

FIG. 7 illustrates an exemplary user interface for a payee to receive a payment transmission, in accordance with some embodiments of the present disclosure.

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In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items or features. Moreover, multiple instances of the same part are designated by a common prefix separated from the instance number by a dash. The drawings are not to scale.

DETAILED DESCRIPTION

The present disclosure generally relates to facilitating digital payments from end-to-end, and more particularly to systems and methods for dynamic generation of messages with unique links to user interfaces for receiving transferred money. In an exemplary embodiment, a customer of a financial institution (also referred to herein as a payer or payor) can transfer money directly to an intended recipient (also referred to herein as a payee) via a hyperlink uniquely associated with the payment transaction. In one example, a payer initiates a payment of a certain value with a certain intended payee. Funds sufficient to cover that value are removed from one or more of the payer's accounts and are maintained in a holding account. A transaction management system generates a hyperlink on an ad hoc basis, and stores that hyperlink in a database. The generated hyperlink is sent to the payee in an email, text message, or other appropriate type of private digital communication. Upon following the link, the payee is presented with a user interface to enter their own debit card (or other account) information, upon which entry the money will be transferred from the holding account to the provided payee account. In the exemplary embodiment, no signup, login/password, biometric, third-party application, or other user-entered authentication is required by the payee, and no entry of account information is strictly required by the payer. In some embodiments, a know your client (KYC) check and/or other forms of security/risk checks may be performed prior to transferring money to the payee.

In some embodiments, the state of the generated hyperlink can be managed to restrict access to the payment. For instance, the hyperlink may be associated with a certain period and/or time limit for access. In other embodiments, the hyperlink may additionally or alternatively be restricted to a certain visit count, by IP address or geographic area, or the like.

In some embodiments, the transaction management system uses its knowledge that a payment was made to instruct the update of the records of other interested entities, without either party to the payment needing to take further action. For instance, there may be an integration with one or more online payment systems (e.g., a property management system) where the transmission of the payment to the recipient triggers an instruction/notification to the online payment system to record the payment to the customer's account. Rather than a pre-configured payment processing system saved in association with the customer account, the system can continuously broadcast out a public stream of payment transaction data regarding payments made through the mechanism described above and herein, and third-party payment system managers can recognize a particular payment from the broadcast as being relevant to their system. That is, the transaction management system could use webhooks to send data (notification of payment) to a plurality of third-party payment systems, where not all the data will be relevant to each third-party system. The third-party system's

receipt and processing of this data may be as simple as updating a ledger of the customer's and/or recipient's account.

In conventional solutions, a customer (payer) and the recipient (payee) both have accounts with a third-party payment system, each requiring a prerequisite sign up process. Further, most payers have a variety of bills to pay with multiple discrete service or product providers, such as rent, individual utilities, subscription services, mortgages, loan repayments, insurance, among many others, potentially requiring sign-up to several completely distinct payment processing systems. Payers may find it burdensome or time consuming to establish and maintain a plurality of accounts. Payees may be subject to various subscription costs or surcharges for use of the third-party systems. Further, setting up these accounts may require the payer and/or the payee to disclose information that they may not want to provide (such as detailed bank account information), over one or multiple third party systems with varying degrees of security.

In contrast to these conventional solutions, the systems and methods described herein empower end users to manage their own digital payments while maintaining security and traceability between payer and payee. Specifically, user interfaces are created and transmitted that provide new mechanisms for people to send and receive money. Because these mechanisms are implemented through already-adopted and relatively universal platforms, such as email and text messaging, payees can receive money without having to download or rely upon third party systems. The systems and methods described herein further mitigate delays in payment transmission that may occur when transmitting money through a middleman, allowing for faster end-to-end payment processing, and removing the requirement for a payer and payee to subscribe to the same third-party service. Still further, the user interfaces described herein may be streamlined to reduce bandwidth and storage requirements otherwise imposed by third party websites and applications. Additionally, the systems and methods described herein can be used in parallel with third party payment processing systems, transmitting notifications of payment so that the payer's/payee's digital payment records can be synchronized across multiple systems. Accordingly, the convenience, speed, security, and accessibility of the computer systems handling digital payment processing and the user interfaces presented to the end users are improved.

FIG. 1 depicts, in accordance with some embodiments of the present disclosure, an environment 100 including a transaction management system 150 for facilitating payment transactions between a payer 110 and a payee 120. As illustrated, environment 100 may include one or more payers 110 (also variously referred to herein as users of a payer device 112, customers, clients, or consumers) who may open an account (e.g., checking or savings account or other banking account) with the owner of a transaction management system 150. In an exemplary embodiment, the entity that owns or manages the transaction management system 150 is a financial entity, such as a consumer banking service 180 for managing the payer's financial account. The consumer banking service 180 may also manage one or more user-facing systems. Payer 110 may, via a dedicated application (app) or website accessed on their mobile device 112 (described in greater detail below with reference to FIG. 2), be a user of such a user-facing system, such that payer 110 can interact with the consumer banking service 180 and transaction management system 150, for example to check or manage account status, make deposits or withdrawals, and the like. The payer 110 may be capable of initiating one

or more payment transactions associated with the payer's bank account, or, in some embodiments, more than one bank account, using their device 112.

As illustrated, the environment 100 may also include one or more payees 120 (also variously referred to herein as users of a payee device 122, recipients (of a payment), or merchants/service providers) who need not be a customer of (or otherwise affiliated with) the consumer banking system 180, but who can, via a device 122, access one or more user interfaces provided by transaction management system 150. A payee 120 may have their own account with a banking entity. In some embodiments, the payee's account is associated with a payment device 124. In an exemplary embodiment, payment device 124 is debit card, however it may be any of, for example, a payment card having a magnetic strip that is swiped in a magnetic reader of a payment reader, a payment device having a Europay/MasterCard/Visa (EMV) chip that is inserted into a corresponding EMV slot of a payment reader (e.g., a secured charge card, credit or debit card, a gift card, a proxy card, etc.), and near field communication (NFC) enabled devices such as a smart phone or EMV card that is tapped at a payment reader and that transmits payment information over a secure wireless connection, or an electronic or virtual account. A payee 120 may be any individual, group, or other entity that the payer 110 wishes to transfer money to for any reason. For example, a payee might be a merchant or service provider that is owed a payment of a particular value from payer 110. The term "money" is used herein simply for convenience, and any asset associated with a value denomination, e.g., cryptocurrency, may apply and be applicable in other embodiments. The term "merchant" may be understood to encompass any business or other entity that sells, leases, or otherwise provides or provided goods or services to payer 110 as part of a financial transaction, an ATM or other device/system with a cash-back function, a bill pay system (e.g., system for sending checks, wires), an automated clearing house (ACH), a party to one or more peer-to-peer payments, or any other entity or representative party or object engaging in a payment transaction with the payer 110. Examples of merchants may be landlords, utility companies, cable companies, banks or other entities from which the payer has taken a loan or mortgage, a subscription service, an online or physical vendor or service provider, an individual known or unknown to the payer 110, or any appropriate party.

Payee bank 130 may include any number of computing servers that manage the payment networks on which the payment device 124 works. The payee bank 130 may receive (and in some embodiments transmit) information or instructions regarding payer 110's payment transactions from transaction management system 150 and/or the consumer banking service 180. For instance, transaction management system 150 may transmit an electronic payment through an automated clearing house (ACH) network from the payer's account(s) with consumer banking service 180 to the payee's account with payee bank 130.

Third party databases 170 may include one or more databases variously storing information regarding the payer 110 and/or the payer 120, and/or their accounts or payment devices. In some embodiments, transaction management system 150 may request information stored at one or more of third party databases 170 to facilitate or assist in a know your client (KYC) check regarding a payment to be transferred, perform a risk analysis of the payee or transaction, the payer 120, and/or the payer device 122, or perform other analytical actions. Third party databases 170 may contain publicly accessible and/or private information (where trans-

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action management system **150** has authorization to access such information), or may contain information owned or managed by transaction management system **150** that is being stored or hosted on a third party-managed server.

The components of environment **100** may communicate with each other over a communication network **160**. Network **160** may comprise one or more network types, such as a wide area network (such as the Internet), a local area network (such as an intranet), a cellular network or another type of wireless network, such as Wi-Fi, Bluetooth, Bluetooth Low Energy, and/or other close-range wireless communications, a wired network, such as fiber optics and Ethernet, or any other such network, or any combination thereof. Although communication network **160** may be any suitable communication network, in one embodiment, communication network **160** is the Internet and payment and transaction information may be communicated in an encrypted format such by a transport layer security (TLS) or secure socket layer (SSL) protocol. In addition, when the network **160** is the Internet, any of the components of environment **100** may use the transmission control protocol/Internet protocol (TCP/IP) for communication.

It will be understood that while, for ease of illustration, FIG. **1** depicts one payer **110**, one payee **120**, one payment device **124**, one payer device **112**, one payee device **122**, one merchant bank **130**, one transaction management system **150**, one consumer banking service **180**, and a plurality of third party databases **170**, environment **100** is not limited to that configuration. In various implementations, any number of payers, payees, banks, services, or devices may be used in any number of configurations. Further, while FIG. **1** illustrates a card or NFC based payment device used by payee **120**, other embodiments may exist where any device or information associated with a financial account may be used in any transaction involving the deduction of funds from that financial account.

In an exemplary embodiment, each of payer device **112** and/or payee device **122** may be, for instance, a respective mobile telephone or smartphone such as an iPhone or Android device, an iPad or tablet device, a laptop or touchscreen device, a PC or stationary computing device, a web appliance, a network router, switch or bridge, or any other practical machine that can communicate via a communication network **160** and is capable of executing instructions (sequential or otherwise) that specify actions to be taken by that device. Payer **110** can use their device **112** (and payee **120** may use device **122**) to access, view, input data, and/or take action in response to delivered content. In an exemplary embodiment, the devices **112**, **122** present information to a user via a display on or connected to the devices **112**, **122**, and take input from the user in relation thereto (for example, through interactive graphics or hyperlinks) via a touchscreen, mouse, keyboard, stylus, or any other appropriate input device. For example, devices **112**, **122** may be capable of receiving and displaying notification data via a dedicated application (app) or website, email, text or instant messaging, voice, SMS, QR code, voicemail, or any other appropriate type of communication.

For ease of reference, the component parts and functions of payer device **112** are described below and herein (e.g., with reference to FIG. **2**) and payee device **122** may be understood to contain the same (or similar) components and to be configured to perform the same (or similar) functions as device **112**. Component parts of a device **122** may be referred to herein by the numerical values of analogous components of device **112**. Some exemplary differences of note between payer device **112** and payee device **122** may be

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described below, however even where no differences are specifically described, the devices **112** and **122** need not be identical to each other in form or function.

FIG. **2** illustrates exemplary components of a payer device **112**, though the device is not limited thereto. The payer device **112** may display a user interface through which a user may interface with transaction management system **150** via communications over network **160**. The user interface may be displayed on device **112** through, for instance, one or more of a web browser, application, text message/alert, device-standard notification, or by any other technique that allows for the display of server-generated content on an electronic display or visual interface **272** of the device **112**. This may be a screen, such as an LED screen, OLED screen, LCD screen, plasma screen, and/or touchscreen (e.g., capacitive or resistive touch display). The visual interface **272** may display user interfaces directly (e.g., on the screen) or indirectly on a surface, window, or the like (e.g., via a visual projection unit). The user interfaces may also allow for input of information via that screen and/or another input device (touch-sensitive components, keyboard, keypad, mouse, trackpad, text, voice call, passbook, wearable or peripheral devices, or any known technique, whether implemented as software or hardware or any combination thereof). Device **112** may in some embodiments include alphanumeric input device **274** (e.g., a keyboard or touch screen keyboard), a cursor control device **276** (e.g., a mouse, a trackball, a joystick, a motion sensor, or other pointing instrument), a signal generation device **278** (e.g., a speaker), a microphone, and/or a static memory **262**, which components also are configured to communicate via local interface **260**, which may include, for example, at least one bus.

Payer device **112** may include a memory **210**. As used herein, memory **210** may refer to any suitable storage medium, either volatile and non-volatile (e.g., RAM, ROM, EPROM, EEPROM, SRAM, flash memory, disks or optical storage, magnetic storage, or any other tangible or non-transitory medium), that stores information that is accessible by a processor **250**. The illustrated embodiment depicts a number of modules or applications stored in memory **210**. These depicted modules may variously represent one or more algorithms, computational models, decision making rules or instructions, or the like implemented as software code or computer-executable instructions (i.e., routines, programs, objects, components, data structures, etc.) that, when executed by one or more processors **260**, program the processor(s) to perform the particular functions of their respective logic. These modules are depicted in FIG. **2** as several discrete components, each labelled as an individual “logic”, however, in various embodiments, the functions of each respective logic may be executable on their own or as part of one or more other modules; that is, any configuration of the depicted logical components may be used, whether implemented by hardware, software, firmware, or any combination thereof.

In some embodiments, memory **210** may store a financial service application **211**, which may be a software application (app) managed by transaction management system **150** or consumer banking service **180** to display and intake information from the user of the device. In some embodiments, memory **210** may store a location determination module **212** (which may be part of financial service app **211**) configured to determine a user’s location. In such an embodiment, payer device **112** may additionally include a location-determination device, such as a device transmitting a beacon or signal, a GPS receiver, or another type of device that may be used by location determination module **212** to

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determine a current location of the customer device. In some embodiments, payer device 112 and/or payee device 122 need not contain either or both of the financial service application module 211 and/or the location determination module 212, and instead may use one or more other modules (e.g., a browser, text messaging application, mail application, or the like) to view and input information as described in greater detail herein.

FIG. 3 depicts an example schematic diagram of certain components of a transaction management system 150. FIG. 3 shows a diagrammatic representation of components of a machine such as a server computer, a client computer, a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a cellular telephone, a smartphone, a web appliance, a network router, switch or bridge, or any machine capable of executing instructions (sequential or otherwise) that specify actions to be taken by that machine. While only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute instructions to perform any one or more of the methodologies discussed herein.

The transaction management system 150 may include a memory 310. As used herein, memory 310 may refer to any suitable storage medium, either volatile and non-volatile (e.g., RAM, ROM, EPROM, EEPROM, SRAM, flash memory, disks or optical storage, magnetic storage, or any other tangible or non-transitory medium), that stores information that is accessible by a processor (e.g., processor 350). Memory 310 may also be understood as a machine-readable medium on which is stored instructions (e.g., software) embodying any one or more of the methodologies or functions described herein. While FIG. 3 illustrates a single discrete memory 310, it will be understood that the embodiments described herein are not limited to any particular arrangement and that other embodiments may store information in one combined memory, or with information stored in a different configuration in one or more memories, some local to the other components illustrated in FIG. 3 and/or some shared with, or geographically located near, other remote computing systems.

As illustrated, a number of modules are stored in memory 310, specifically, user input module 331, link generation module 332, payment risk module 333, notification module 334, control logic 321, and communication logic 322. These depicted modules may variously represent one or more algorithms, computational models, decision making rules or instructions, or the like implemented as software code or computer-executable instructions (i.e., routines, programs, objects, components, data structures, etc.) that, when executed by one or more processors 350, program the processor(s) to perform the particular functions of their respective logic. These modules are depicted in FIG. 3 as several discrete components, each labelled as an individual “module” or “logic”, however, in various embodiments, the functions of each respective logic may be executable on their own or as part of one or more other modules; that is, any configuration of the depicted logical components may be used, whether implemented by hardware, software, firmware, or any combination thereof. Further, the modules need not necessarily be part of transaction management system 150 and may instead be distributed over one or more computing systems. The capabilities of these various logics are described in greater detail below.

The transaction management system 150 may include control logic 321, including one or more algorithms or models for generally controlling the operation of the trans-

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action management system 150. The memory 310 may also, in one embodiment, include communication logic 322, including one or more algorithms or models for obtaining information from or communicating information via network 160 (FIG. 1). The transaction management system 150 may, via communication interface 365, operate to exchange data with various components, systems, and/or devices on the network 160 or any other network. For instance, communication interface 365 and communication logic 322 may be used (by, e.g., any of modules 331-334) to access data from or send data to any of one or more payer devices 112, payee devices 122, payee bank 130, one or more external or third party processing systems 140, one or more external or third-party databases 170, one or more web servers 370, and/or one or more payment processing systems 380. In some embodiments, communication logic 322 may use APIs provided by these entities or webhooks established between the transaction management system 150 and/or one or more third party payment processing systems 140 (or other entities) to obtain their respectively stored data or transmit data or instructions to their systems. However, other methods of data collection/transmission may alternatively be used such as one or more software development kits, which may include, e.g., one or more application programming interfaces (APIs), web APIs, tools to communicate with embedded systems, or any other appropriate implementation.

In the embodiment of FIG. 3, one or both of web server 370 and payment processing system 380 may be owned or managed by a single entity, such as consumer banking service 180. In some embodiments, consumer banking service 180 may also own or manage transaction management system 150, such that payment processing system 380 can be used to move, deduct, and/or process funds from an account at consumer banking service 180 associated with the payer 110, based on decisions made and instructions issued by transaction management system 150. Information or notifications regarding the transaction(s), and/or the processing of various account(s) may be displayed to the payer 110 or payee 120 via one or more user interfaces respectively transmitted to payer device 112 or payee device 122 via web server 370. In an exemplary embodiment, web server 370 can deliver, to payer device 112, one or more of various user interferences that provides the payer with the ability to perform one or more banking-related functions, which may include any or all of: making payments from one or more of payer's accounts with consumer banking service 180, enrolling in one or more financial accounts, inputting account settings, management of financial accounts (e.g., checking/savings accounts), viewing of account statements or metrics, receiving financial alerts/notifications, making deposits/withdrawals, direct deposit, bill pay, money transfer, check deposit, and/or any other relevant activities. In an exemplary embodiment, web server 370 can deliver, to payee device 122, one or more of various user interferences that provide the payee with the ability to enter bank account information (e.g., debit card information) that can be used by system 150 to transfer a monetary value based on a payment instructed by payer 110. In some embodiments, the data presented to the payer device 112 may be generated based on data stored in user profile database 341 and/or transaction database 342, and the data presented to the payee device 122 may be generated based on data stored in the transaction database 342 and/or hyperlink database 343, however, in other embodiments, such data may be collected from one or more databases on system 150 or an external system.

While communication logic 322 is illustrated as being a discrete logical component, in an alternative embodiment,

the transaction management system **150** may include communication logic **322** as part of any of modules **331-334** or control logic **321**. In another alternative embodiment, the communication logic **322** may communicate with third-party systems and/or may coordinate with the control logic **321** to read or write data to memory **310** or to another data repository (not shown) within the transaction management system **150**.

The logics of the exemplary transaction management system **150** depicted in FIG. **3** may be executed by one or more processors **350**, which may include any of (or any combination of) a central processing unit (CPU), graphics processing unit (GPU), digital signal processor (DSP), application specific integrated circuits (ASICs), radio-frequency integrated circuits (RFICs), other specialized processor or combination of processors, or other circuitry that communicates to and drives the other elements within the transaction management system **150** via a local interface **360**, which can include at least one bus. While FIG. **3** illustrates one processor **350** that implements all of the various logics and modules in the transaction management system **150**, it is possible in other embodiments for the system **150** to employ multiple processors. In one such alternate embodiment, discrete processing elements may be used for each of (or any subset of) modules **331-334**, control logic **321**, and communication logic **322**, or any portions or subsets of those logics. In some embodiments, the processing of system **150** is not limited to being performed by a processing element connected to the local interface **360**, but instead, any portion of processing in support of the various logics may be distributed over one or more computer systems that may be remotely located. In some embodiments, transaction management system **150** may be implemented in whole or in part as a machine learning system (e.g., neural network software) for achieving the functionalities described herein. In one embodiment, one or more of modules **331-334** (or any subset of any of those logics) may be implemented at least in part as one or more machine learning algorithms.

Memory **310** may be configured, in some embodiments, to include various databases. While the term “database” or “repository” is used with reference to elements **341**, **342**, **343**, **344**, and **170** these components are not so limited nor is any particular form or configuration of data storage mandated, and the described “databases” may alternatively be one or more of an indexed table, a keyed mapping, or any other appropriate data structure, or any combination thereof.

Memory **310** may be configured, in some embodiments, to include a user profile database **341** that stores account information related to one or more consumer accounts with a single financial entity (e.g., a consumer banking service **180**) that owns or manages system **150**, or accounts with other financial entities linked to such consumer accounts. An account may generally be understood to be associated with a payer **110**, however the association need not be 1:1, such that an account may have multiple owners (joint account) or a payer **110** may have multiple accounts of any of various types (e.g., checking, savings, certificates of deposit, money market accounts, individual retirement accounts, and the like). In an exemplary embodiment, the accounts are those from which money can be easily transferred, that is, traditional checking and savings accounts, though they are not limited thereto. In some embodiments, user profile database **341** may include one or more tables, each entry in the table(s) corresponding to a unique set of account information or a unique user. In some embodiments, user profile database **341** may include information about a payer and/or a related or associated account, such as a unique account ID,

an associated user ID (or customer ID), a current account balance, information regarding pending deposit/withdrawal activity if such information exists (e.g., whether money is currently being held in any holding accounts and the associated information), user name, contact information (e.g., email address, mailing address, telephone number, etc.), date of birth, payment card information (e.g., payment card number, expiration date, cardholder name, security code) for each of any number of associated payment cards, customer account settings and/or preferences, and the like.

User profile database **341** may also include real-time and/or historic information relating to any of: customer account creation/removal/edits, deposit/withdrawal activity (e.g., dates of activity, amount of deposit/withdrawal) on customer accounts, activity by any users that may have shared or joint accounts with the customer, customer financial health metrics or scores, purchases made or pending using cards tied to any of the customer's accounts, customer activity on one or more user interfaces delivered by web server **370**, and/or metrics generated on any of the above such information, whether based on an individual account or user, or in aggregate. In some embodiments, as various users login, make edits, changes, view, search, click, interact with, or otherwise engage in activities with the user interface on system **150**, such activity may be stored in database **341** as indicators that may be considered, individually or in aggregate, in an analysis of payer **110**'s activity. In some embodiments, this information may be understood as user engagement metrics or behavioral biometrics.

Memory **310** may also store a transaction database **342** that contains information regarding individual payment transactions initiated by users identified in user profile database **341**, whether pending or completed. Transaction database **342** may, in some embodiments, store information detailing every transaction made or requested in connection with a user's account(s), stored in association with a unique transaction ID and information sufficient to uniquely identify the associated user ID in database **341** (and/or their relevant financial account information). The data in database **342** may include any of approved transactions, rejected transactions, withdrawn transactions, pending transactions, refunded transactions, and the like, along with account balances resulting from such transactions. Transaction database **342** may also store, in association with a transaction, information sufficient to identify any payment recipients or entities that have transferred money into the payer's accounts. In some embodiments, a payment recipient may be identified by a name or address information, or account information, for example where the user has previously made payments to the payee recipient or has saved the recipient's information in system **150** (via financial service application **211**, web browser, or other interface). In an exemplary embodiment, database **342** contains, in association with each stored transaction, contact information for the payment recipient such as an email address or phone number, or other appropriate contact information by which transaction management system **150** can transmit one or more digital messages to the recipient. In some embodiments, a transaction may be stored in association with one or more values related to a state of the transaction, such as whether it is active or expired, whether it has been approved or denied, whether it is pending, an expiration date/time, or the like.

Memory **310** may also store a hyperlink database **343**, containing at least one hyperlink value stored in association with information sufficient to uniquely identify a transaction ID stored in transaction database **342**. In an exemplary

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embodiment, a hyperlink is unique to a transaction, that is they are stored in a 1:1 relationship. In some embodiments, a hyperlink may be stored in association with one or more values related to a state of the hyperlink, e.g., a count number of access attempts (wherein access may be restricted if the count exceeds a certain value, or drops to zero, in embodiments where the count is added to or subtracted from upon an access attempt, respectively), an access date/time, etc. In some embodiments, a hyperlink may be stored in association with an expiration date/time or timeout value, indicating that the hyperlink is only accessible for a finite period of time. In some embodiments, a hyperlink may be stored in association with one or more rules that, when implemented, restrict or exclude access to the URL specified by the hyperlink based on e.g., IP address, device ID (MAC address), geographic region, whether the visitor is using a VPN, and/or other factors as appropriate. In some embodiments, the state of the hyperlink may indicate or be associated with the state of the transaction (e.g., if the hyperlink has/has not expired, the transaction has/has not expired), or vice versa.

Memory 310 may also store a third-party payment system 344 that stores information related to one or more third party payment systems, such as property management systems, utility management systems, and the like. In an exemplary embodiment, the transaction management system 150 may automatically transmit notification messages to one or more systems in third-party payment system 344 when payment transactions are completed and processed between a payer 110 and a payee 120. In an exemplary embodiment, this is done through a broadcast message to all the systems in database 344 or a public broadcast. In other embodiments, payment systems in database 344 may be stored in association with a category (e.g., property management) or other qualifier (e.g., location (city/state)). In such a case, transaction management system 150 may only transmit messages to a subset of systems in database 344, for example, those belonging to a certain category or sharing a certain qualifier.

In an exemplary embodiment, system 150 may update information in databases 341 and/or 342 based on data collected from web server 370. In the embodiment of FIG. 3, web server 370 is remote from system 150 and therefore such information is collected via communication logic 322. In other embodiments, web server 370 may be local to, or integral with, system 150 and such data may be accessed over local interface 360. Web server 370 may transmit information to and receive data from user device 112.

In an alternate embodiment, databases 341, 342, 343, and 344 are not stored in transaction management system 150 itself, but are remotely stored in a shared memory (not specifically shown) so as to be accessible via one or more communication networks, such as network(s) 160. In some embodiments, the databases 341, 342, 343, and 344 may be accessible over the Internet and payment and other information may be communicated between system components in an encrypted format such by a transport layer security (TLS) or secure socket layer (SSL) protocol. In such an embodiment, both transaction management system 150 and payment processing system 380 are owned or managed by consumer banking service 180, and both systems 150 and 380 may access shared data. By these means, as either of systems 150 or 380 updates data in databases 341, 342, 343, and 344, the updated information is available to the other system.

FIG. 4 illustrates an exemplary process 400 by which transaction management system 150 may dynamically generate and transfer messages with unique links to user inter-

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faces for receiving transferred money. In step 402, user input module 331 receives a transmission from financial service application 211 (or another component of payer device 112). This transmission contains data collected from payer 110 via a user interface displayed on payer device 112, the data including at least information sufficient to identify a payee and a payment amount. User input module 331 is configured to store, in user profile database 341 and transaction database 342, data associated with this initiated transaction.

In an exemplary embodiment, transaction management system 150 transmits, to financial service application 211, a user interface through which the payer can transfer money. This UI may be displayed to the payer through a dedicated app or browser on device 112, or through any interface through which information can be input via text, touch, or voice input, or any other input mechanism, and transmitted to the system 150. In some embodiments, financial service application 211 may interface with one or more other operating system or third party applications to facilitate user input (e.g., personal assistant apps, accessibility apps, or the like).

FIG. 5 illustrates an exemplary user interface 500 displayed by financial service application 211 on the payer device 112 that may be used to facilitate a user's input of their payment information. User interface 500 includes payee selection element 510, a user interface element capable of taking in a value associated with an intended recipient of a money transfer. While element 510 is depicted as a drop-down menu, other embodiments may implement element 510 as a text field, a predictive text field, an audio input field, or the like. In FIG. 5, the value illustrated in element 510 as being selected is a "one time payment," upon the selection of which the payer would be prompted to enter contact information for the payee, such as an email address or phone number (to which an SMS or text message may be sent). Other illustrated values include a variety of previously stored/saved payees (previously stored by the payer 110), such as landlord, electrical or utility company, an individual (here "John Smith"), or a field to enter and save a new payee. These saved values may be preferred in embodiments where a payer makes multiple or recurring payments to the same payees, and wishes to store their contact information to avoid duplicated effort (or the possibility of mistake/error) in reentering payee information. FIG. 5 also illustrates a field 512 through which the payer can enter an amount to be paid.

In some embodiments, the UI 500 need not require that the user specifies any payee information, that is, the payer may simply initiate a generic transaction for a certain value specified in field 512, and may leave field 510 empty (or field 510 may not be displayed to the payer at all). Such an embodiment may be useful when the payer intends to communicate with the payee directly to transmit to them a payment link (described below).

Some embodiments may also include a field 514 in which the user can input a "description" to provide context for the payment. In some embodiments, this may be a freeform character string, and in other embodiments, it may be a selected value, e.g., from a variety of pre-set categories. Still other embodiments may contain a field (not shown) allowing the payer 110 to specify or select a financial account (associated with the payee in database 341) from which the payer intends the money for the transaction to be withdrawn or debited. In some embodiments, the UI 500 may also contain a field allowing the payer 110 to specify a third-party account number (e.g., a service account number) or other identifier (address, ID number, etc.). For instance, in a case that the payer is paying their electrical bill, the payer may be

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assigned an account number with their electrical company, and may enter the same for the payment transaction to be associated with that account, so that the payment may be later recognized and properly recorded by the electrical company's third party software **140** (or in other embodiments, e.g., a property management software, rent payment software, etc). Alternatively, such information may be specified in field **514**. Of course, the fields and values shown in the illustrated UI **500** are merely exemplary, and any values may be possible in different embodiments.

Embodiments may also exist where the payer does not have to access UI **500** (or similar) to instantiate a payment, and instead, a payment may be initiated by transaction management system **150** on a recurring or preset schedule based on preference or setting information entered by the user via financial service application **211**. For instance, in the case of a recurring payment (such as rent), the payer may preset a recurring date/time of payment (e.g., monthly) and payment recipient (e.g., stored contact information for a landlord), and transaction management software **150** may instantiate a new transaction and store the same in database **342** at the scheduled time. In some embodiments, the payer may specify, in the initiation of the payment transaction (step **402**), a future date/time at which the payment should be made (also referred to herein as a scheduled payment), such that funds are not withdrawn from the payer's account(s) until the specified date/time.

In the exemplary embodiment of FIG. 5, the payer **110** does not need to enter user/account information specifically to authenticate this transaction. In some embodiments, the payer may also be logged into financial services application **211** or a website via a browser app, and therefore, user input module **331** may obtain information regarding the payer from cached data or session data. This information may be sufficient to identify the information associated with the payer in user profile database **341** (e.g., name and/or account information, account balance, etc.). In some embodiments, user input module **331** may additionally or alternatively collect data about the payer from other information collected from the payer device **112**, such as MAC address, IP address, device ID, location information or the like, and may use that information to identify the user profile information (in database **341**) associated with the payer and/or authenticate the payer device. In still other embodiments, the user interface **500** may contain one or more fields for user authentication, or may require additional authorization once information has been entered on screen **500**, such as password information, biometric information, confirmation of a transmitted code or two-factor authentication, or the like. This additional authorization information may additionally or alternately be used in some embodiments by user input module **331** to identify the appropriate user information and user accounts in database **341**.

In response to the payer's instantiation of a payment, in step **404**, transaction management system **150** (in some embodiments, link generation module **332**) determines whether the payer's account has sufficient funds to cover the payment transaction, that is, whether the balance of the payer's account meets or exceeds the value entered by the payer. In some embodiments, the transaction management system **150** may check only the account specified by the payer (if any) or a default account (e.g., checking account) of the payer, in which case, if the balance falls short of the specified value (NO in step **404**), the process proceeds to step **424** and a payment denial is transmit to the payer device **110**. The payer may be notified of such denial through a

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display on their device **110** (step **426**), or another message (e.g., text, email, voice call or message, etc.) appropriate.

In some embodiments, rather than a single specified or default account, transaction management system **150** may consider the payer's accounts holistically. For instance, the transaction management system **150** may obtain account information for all accounts belonging to the payer from user profile database **341** and may determine based on the respective balances of those accounts, withdrawal restrictions, and/or user preferences whether or not funds can be withdrawn from a plurality of accounts to fund the payment transaction. In some embodiments, transaction management system **150** may employ one or more machine learning algorithms to determine whether splitting the funding over multiple accounts is recommended or possible, the output of which algorithm(s) is a binary determination of whether sufficient funds are available to the payer at the time the transaction is initiated. In still other embodiments, the payer may have indicated that they wished to make a scheduled payment at a future date, in which case transaction management system **150** may make the determinations of step **404** only at the specified scheduled time rather than at the time the transaction is first input by the payer **110**. If the accounts, taken together, are still determined to contain insufficient funds (or insufficient accessible funds), the process proceeds to steps **424** and **426** as described above.

In some embodiments, the transaction management system **150** may not consider withdrawal from a plurality of accounts without approval from the payer, whether set in advance or received at the time of the transaction. For instance, the system **150** may send a notification to the payer device **110** that the specified (or default) account contains insufficient funds and may prompt the user as to whether to access a different account or to cancel the transaction.

If the payer's account(s) contains sufficient funds (YES in step **404**), the process proceeds to step **406**, in which the amount specified by the user in step **402** is removed from the payer's account(s) and placed in a holding account managed by the transaction management system **150** and/or the consumer banking service **180**. The holding account may be thought of as an intermediate account between the payer and the payee. The holding account is used so that the payer cannot double spend the money intended for transmission to the payee. In addition, an entry for the transaction is stored in database **342**, based on the information provided in step **402**.

In step **408**, link generation module **332** generates a hyperlink based on the input information obtained in step **402** and stored in transaction database **342**. The generated hyperlink is stored as a new entry in hyperlink database **343**. The hyperlink is unique to the transaction, such that a person who follows the hyperlink will arrive at a user interface associated solely with the transaction initiated in step **402**, regardless of what if any other transactions the payer **110** may have initiated or the payee **120** may have been sent.

The hyperlink may be sent to the payee by the transaction management system **150** or by the payer themselves. For instance, the system **150** may transmit to the payer (or the payer may request via application **211**), the generated hyperlink. The hyperlink is received by the payer in step **410-a**, and the payer may then in turn send the hyperlink to the payee in step **410-b**, for example by copying/pasting the hyperlink into an email or text message between the payer device **110** and the payee device **120**. FIG. 6A illustrates an exemplary user interface **600** where a payer has transmitted a text message **602** to a payee (here, "John") containing the hyperlink **605** generated in step **408**. The hyperlink may take

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any format, however in the exemplary embodiment is a clickable or selectable interface (regardless of the text displayed on the payee device **122**), which the payee can follow to access one or more UIs. In such an embodiment, the payer doesn't have to enter any contact information for the payee into the transaction processing system **150**, or wait for the transaction processing system **150** to issue a formal notification to the payee. Rather, the payer can text or send a message to a payee (e.g., their landlord), from a source known and recognizable to the payee (payer's phone number or email) and at the convenience of both parties. Alternatively, in step **412**, the transaction processing system **150** may transmit the hyperlink to the payee device **122** via an electronic message. In some such embodiments, the payer may provide contact information for the payee to the system **150** or such information may already be accessible in a memory.

The message with the hyperlink (whether sent to the payer or payee) may take the form of an email or text message, an SMS (short message service) message, a QR (Quick Response) code, as a notification (e.g., a push notification) or alert to the devices **112** or **122** through one or more apps or operating system messages, e.g., an app-based notification, a phone call, voicemail, a sound or visually-based alert or ping, or any other appropriate means. The message may be, for instance, a human-parseable or understandable message (such as a character string, video, or audio recording) that includes the generated hyperlink and/or details of the transaction.

FIG. **6B** illustrates an exemplary user interface **620** where system **150** has transmitted a message to payee device **122**. In the illustrated embodiment, the message is a text message **622** containing information about the payer name and a description of the transaction, based on the information collected in step **402** and stored in transaction database **342** and/or user profile database **341**. The hyperlink **625** is similar to that described above with reference to FIG. **6A**. In some embodiments, the text message **622** may identify itself to be from a bank (e.g., consumer banking service **180**). Of course, the messages illustrated in FIGS. **6A** and **6B** are merely exemplary and other embodiments may differ in form or content.

In step **414**, the payee follows the hyperlink in the message using their device **120** by clicking on, touching, tapping, copying/pasting, or otherwise appropriately interacting with the hyperlink. By following the hyperlink, the user input module **331** delivers to the payee device **122** one or more user interfaces, usually via a browser application but in some embodiments via one or more other applications such as a banking application displayed on the device **122**, or device-specific applications. The user interfaces contain one or more fields for the payee to enter debit card information (e.g., the 16 digit value), expiration date, etc., or other information (e.g., bank account number, routing number) sufficient to facilitate a credit or payment to the identified debit card/account. In some embodiments, the user interface may also display information describing or relevant to the transaction, such as the payer name, the amount of the transaction, a description (e.g., as entered in element **514** of FIG. **5**), or the like. In some embodiments, this information may be displayed to the payee on a screen or user interface separate from and/or preliminary to the user interface through which the payee can enter their debit card information. FIG. **7** illustrates an exemplary user interface **700** for a payee to receive a payment transmission. A payer name and transaction description (optional) is displayed in field **702**, and debit card information can be entered in fields

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710, **712**, and **714**. An "Accept Payment" button triggers the transfer of funds from the holding account to the entered destination account (in some embodiments, first requiring a pre-requisite risk analysis and decision and described herein). Of course, the illustration of FIG. **7** is merely exemplary, and the form and content of the UI may vary in other embodiments.

Prior to displaying the user interface(s) to the payee, the user input module **331** may refer to information in the transaction database **342** and/or hyperlink database **342** to determine whether or not the hyperlink for the transaction is still valid and therefore still useable for the payee to complete the transfer of funds. For instance, user input module **331** may refer to state information stored in association with the hyperlink. In one embodiment, the hyperlink is associated with an expiration time, either a date/time at which the link becomes inaccessible (e.g., January 1st at 12:00 am), or a "timer" to expiration (e.g., 1 hour, 24 hours, 2 weeks, or any appropriate value). In the case that the expiration is a countdown, a time difference between the date/time the request was initiated by the payer (or alternatively, when the request was processed/approved by the system **150**, or when the hyperlink was generated) and the current time the hyperlink is accessed by the payee is calculated, and that time difference is compared to the timer value. If the actual elapsed time exceeds the stored timer value, the hyperlink is considered to have expired, and the payee is instead directed to an error screen, a default screen, an explanatory screen, or the like. In some embodiments, the money transferred from the payer's account to the holding account may be credited back to the payer's account from which it was initially debited at or after the time that the hyperlink has expired. In other embodiments, the "expiration" or the hyperlink may be triggered by a number of visits or hits to the user interface (e.g., website) specified by the hyperlink. For example, a counter value is stored in database **343**, and is altered in accordance with the number of times the URL of the hyperlink is used to request data from the web server. In one implementation, the counter begins at zero and is incremented in the memory upon each hit until a maximum or preset number of hits is reached. In another implementation, the counter is initially set to a non-zero value, and is decremented in the memory upon each hit until it reach zero. As one example, the counter may be capped at a value of 1, so that the payee only has one opportunity to obtain the transferred funds.

In Step **414**, the recipient uses the user interface to enter their debit card information to receive the transfer. In such an embodiment, no login or signup or download of an app is needed, and no account must be created beyond the financial institution account that the payee is already in possession of. Once that information has been entered, the payee confirms their information (field **720** in FIG. **7**) to initiate transfer from the holding account to the payee's account as specified in the user interface **700**.

It is possible that the payer may have mis-entered the payee information, or that the payee may be a fraudulent or bad actor. In such scenarios, the transaction management system **150** may wish to avoid finalizing the transfer of funds. Accordingly, in step **416**, payment risk module **333** performs a risk analysis to determine a level of risk associated with the transaction. In one embodiment, the risk analysis may be a KYC check (and/or other risk/security checks) on person receiving money. Because the payee does not necessarily go through a formal onboarding process, the KYC check is performed in real-time at the point that credit of the transferred funds is request. The KYC check and/or

other risk analyses may be performed in some embodiments via one or more machine learning or other algorithm models using information collected from the payee device such as such as MAC address, IP address, device ID, location information or the like. Payee device **122** may have a location determining module **212** that transferred location information of the payee device to the payment risk module **333**. In some embodiments, payment risk module **333** may consider security information from payer device **112** or payee device **122**, such as security logs or known flaws or vulnerabilities, virus check results, malware, intrusion attempts, and the like. The risk analysis models implemented by payment risk module **333** may consider, for instance, a payee's IP address, the browser URL string followed, known information about the payee, and the like, to identify a level of risk in the payee. If the risk value exceeds a risk value, the transaction is cancelled (along the lines of step **424**). In some embodiments, payment risk module **333** may also consider information in transaction database **342** or user profile database **341**, such as whether the payer has made authorized payments to this payee in the past (and the conditions and details of those payments), whether multiple payments have been authorized at once, whether additional payee information is available, and so on. The payment risk module **333** may also access data from third party databases **170** in its analysis, e.g., the reputation of the payee and/or the financial institution associated with the payee's provided debit card information (i.e., who the issuing bank is), previous transaction information or illegal activity associated with the provided debit card (i.e., whether it may be a stolen card), or any other available third-party data.

In step **418**, the payment risk module **333**, having determined based on its analysis that the risk is sufficiently low (that is, below a threshold), instructs that money be transferred from the holding account (stored in transaction database **342**) to the debit card input by the payee in step **414**. In the exemplary embodiment, this money transfer is performed through any appropriate real-time mechanism for person-to-person (P2P) or bank-to-bank payments (such as, e.g., Visa Direct or another real-time payments network). The payment transaction between the payer and the payee is thereby concluded, and any relevant information regarding the closed status of the transaction (or other relevant information, such as risk analysis data) may be stored in association with the transaction in transaction database **342**.

Transaction management system **150** may send confirmation notifications (e.g., transmission **420**) to one or both of payer **110** and payee **120** (step **422**). In some embodiments, transaction management system **150** may also, in response to the transmission of funds made to the payee's debit card, trigger a notification to one or more third party payment processing systems. As just one example, a transaction from payer to payee may be a rent payment from a tenant to a landlord, and the third party payment system that might otherwise accept the payer's account information (absent the performance of the process **400** described herein) may be a property management software. This notification is performed after the money has already moved from the payer to the payee; that is, the notification is simply an informative message and/or a request for the third party system to update its own ledgers or records to conform with or otherwise be in line with the payments that have been facilitated. In an exemplary embodiment, the notifications are not targeted communications from a single source to a single destination, but rather, are broadcast communications sent out by the system **150**. For instance, system **150** may send out a

broadcast stream of transaction information, each broadcast communication being automatically triggered by the credit of a desired amount of money to a payee account. This process might involve, for instance, the generation of serialized data representative of the information regarding the transaction (such as any of information sufficient to identify a payer and/or a payee, an account number, a payment amount, etc.), or from which such information may be derived. The broadcast communication may be made on a transmission to each of the plurality of webhook URLs, or an HTTP command (e.g., HTTP GET request) containing the generated serialized data, though it is not so limited.

In one example implementation, the system **150** establishes, at a point in time in advance of step **430**, one or more webhook interfaces with different third party payment processing systems, such as property management software, and the like. The contact information and/or webhook information for these system may be stored in some embodiments in third party payment system database **344**, with each of a plurality of webhook URLs being respectively associated with each of the plurality of remote servers/systems specified in the database **344**. These third party systems may be include, for instance, the most common or most well-known/widely-used systems, may include systems relevant or specific to a transaction type or category, may be specified by the payer **110**, and/or may be otherwise collected or selected in any appropriate way. The payer is assumed to use one of the pieces of third party software; if the payer does not use one of the payment process software configured to listen to the broadcast message, the transaction itself is unimpacted (as it is already complete). However, embodiments may be possible where one or both of the payer and payee have not performed any sort of preauthorization/pre-signup/pre-configuration with a third party payment processing systems, at least of the type of payment processed in steps **402-422**.

In step **432**, the third party property management system **140** monitors the public stream of payments transmitted by the transaction management system **150** in step **430**. When the third party property management system **140** finds a "match" (that is, a conformance) between the information specified in a broadcast communication and one or more of their internal property management records (step **434**) the system **140** may record the transaction. For instance, they may add a record of the transaction to their internally stored ledger(s), which may take the form of a table, database, or other appropriate data storage (step **436**). In some embodiments, they could notify transaction management system **150** and/or consumer banking service **180** that they have updated their records (not specifically shown).

Conventionally, a customer must log into a plurality of different third party payment processing systems (e.g., property management systems, utilities payment systems), and enter their own debit card number into each disparate system in order to pay the multiple bills they are obligated to pay. This exposes the customer's payment information to unknown parties across potentially insecure interfaces rather than trusted entities. However, the systems and methods described above with reference to FIGS. 1-7 allow a banking system to provide to its user in real-time unique payment mechanisms facilitating payment directly between a payer's and a payee's devices, without the need for any intermediate payment systems. The conventional system is thereby inverted; in the systems described herein, the payee may collect via their debit card rather than the payer entering their debit card to send money. When a payee launches a website using the hyperlink sent to them, they do not have to log into or sign up with any third party systems or

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download any apps to their device. The banking system may additionally, through continuous notifications to third party payment systems, send instructions such that the ledgers of the third party payment systems align with the directly facilitated payments. The systems and methods described herein provide several technical advantages and benefits over conventional solutions. Unlike conventional “recommendations” made by human tellers or advisors at banks, a human does not need to intercede or make any deliberated decision for a payment to be processed. Further, the payer and payee need not subscribe to the same service or system, allowing electronic payments to be transmitted and received without the creation of new or multiple accounts, allowing for faster and more robust transactions. Further, because fewer computers systems are involved, the overall technical environment is simplified, such that data transmission and/or security can be better managed. Further still, because text or email or other messaging may be used from an entity known to the payee (the payer), both parties can feel confident in the source of the transaction, even where the payee is unfamiliar with the payer’s bank.

The foregoing is merely illustrative of the principles of this disclosure and various modifications may be made by those skilled in the art without departing from the scope of this disclosure. The above described embodiments are presented for purposes of illustration and not of limitation. The present disclosure also can take many forms other than those explicitly described herein. Accordingly, it is emphasized that this disclosure is not limited to the explicitly disclosed methods, systems, and apparatuses, but is intended to include variations to and modifications thereof, which are within the spirit of the following claims.

As a further example, variations of apparatus or process parameters (e.g., dimensions, configurations, components, process step order, etc.) may be made to further optimize the provided structures, devices and methods, as shown and described herein. In any event, the structures and devices, as well as the associated methods, described herein have many applications. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein, but rather should be construed in breadth and scope in accordance with the appended claims.

What is claimed is:

1. A transaction management system for managing direct payer to payee transactions without intermediate payment systems, the transaction management system comprising:

a server having an electronic communications interface capable of receiving data from a payer computing device and a payee computing device via a digital communications network, wherein the server is associated with and is in electronic communication with a trusted consumer banking service computing device associated with a consumer banking service where a user associated with the payer computing device holds a first financial account;

wherein an application is installed on the payer computing device, and the application is configured to generate one or more user payment interfaces on the payer computing device which allow a payer to enter electronic payment transaction information;

wherein the payer computing device interacts electronically with the server and the trusted consumer banking service computing device, wherein the payee computing device interacts electronically with the server, and wherein the server interacts electronically with one or more third party payment processing computer systems to record payments made by payers to payees, wherein

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recording payments updates a ledger representing one or more accounts held by a payer or a payee with the one or more third party payment processing computer systems; and

the server, comprising a memory configured to store instructions and at least one processor configured to execute the instructions to perform operations comprising:

receiving, by the server and from the payer computing device, an electronic payment transaction comprising a monetary value;

in response to identifying, by the server, the first financial account from which to debit the monetary value for the electronic payment transaction and based on receiving the electronic payment transaction, debiting the first financial account of the monetary value;

crediting, by the server, a holding account established at the server and associated solely with the payee for a credit amount, wherein the holding account is separate from the first financial account, wherein the credit amount is the monetary value for the electronic payment transaction;

generating, by the server, a unique hyperlink associated solely with the electronic payment transaction;

transmitting, by the server or the payer computing device, the unique hyperlink via an electronic communication to the payee computing device, wherein accessing the unique hyperlink on the payee computing device will cause the payee computing device to generate a payment acceptance interface uniquely associated with the electronic payment transaction for entry by the payee via the payee computing device of debit card information which is associated with a second financial account associated with the payee and to which the transaction management system can credit the monetary value to complete the electronic payment transaction, wherein the payment acceptance interface is generated on the payee computing device via a browser or a device-specific application already native to the payee computing device;

receiving, by the server and in response to the payee computing device accessing the unique hyperlink, a request to deliver the payment acceptance interface associated with the electronic payment transaction to the payee computing device;

in response to determining that the unique hyperlink is valid, delivering, by the server, the payment acceptance interface associated with the electronic payment transaction to the payee computing device;

receiving, by the server and based on payee interaction with the payment acceptance interface on the payee computing device, the debit card information associated with the second financial account associated with the payee computing device, wherein the payee interaction occurs without a requirement to create an account with the transaction management system, or log in to the transaction management system, or download any dedicated application associated with the electronic payment transaction;

based on receiving the debit card information, processing the electronic payment transaction by transmitting, by the server, a debit instruction to the holding account to debit the holding account for the monetary value, and a credit instruction to a second financial institution associated with the second finan-

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cial account to post a credit to the second financial account for the monetary value; and
 automatically, in response to processing the electronic payment transaction, transmitting, by the server and to the one or more third party payment processing computer systems via an electronic broadcast communication, payment transaction data between the payer and the payee.

2. The transaction management system of claim 1, wherein transmitting the unique hyperlink via the electronic communication to the payee computing device comprises providing, for display via a message graphical user interface for viewing the electronic communication, the unique hyperlink together with a text message element for generating a message to the payee computing device.

3. The transaction management system of claim 1, wherein receiving, by the server and from the payer computing device, the electronic payment transaction comprising the monetary value comprises:

providing, for display via a user interface of the application installed on the payer computing device, a payee selection element and a transaction value element; and identifying the monetary value and the payee computing device based on user interaction with the payee selection element and the transaction value element.

4. The transaction management system of claim 1, wherein the operations further comprise debiting the first financial account of the monetary value based on determining that the first financial account or one or more additional financial accounts corresponding to the payer computing device have sufficient funds to cover the electronic payment transaction.

5. The transaction management system of claim 4, wherein the operations further comprise generating, utilizing a machine learning model, a feasibility recommendation that the first financial account or the one or more additional financial accounts corresponding to the payer have sufficient funds to cover the electronic payment transaction.

6. The transaction management system of claim 1, wherein delivering, by the server, the payment acceptance interface associated with the electronic payment transaction to the payee computing device comprises providing, for display:

a receive payment message indicating the monetary value and a payer identifier corresponding to the payer computing device, and
 one or more debit card information entry fields.

7. The transaction management system of claim 1, wherein generating, by the server, the unique hyperlink associated with the electronic payment transaction comprises generating a hyperlink entry for the unique hyperlink within a hyperlink database.

8. The transaction management system of claim 7, wherein determining that the unique hyperlink is valid comprises, in response to determining that the payee computing device has accessed the unique hyperlink, accessing the hyperlink entry within the hyperlink database to verify that the unique hyperlink is valid.

9. The transaction management system of claim 1, wherein the operations further comprise:

determining an expiration trigger associated with the unique hyperlink; and
 determining that the unique hyperlink is valid by determining that the expiration trigger has not occurred.

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10. The transaction management system of claim 9, wherein the expiration trigger comprises a number of visits to the payment acceptance interface corresponding to the unique hyperlink.

11. A computer-implemented transaction management method for managing direct payer to payee transactions without intermediate payment systems, wherein a server comprises an electronic communications interface capable of receiving data from a payer computing device and a payee computing device via a digital communications network, wherein the server is associated with and is in electronic communication with a trusted consumer banking service computing device associated with a consumer banking service where a user associated with the payer computing device holds a first financial account;

wherein an application is installed on the payer computing device, and the application is configured to generate one or more payment interfaces on the payer computing device which allow a payer to enter electronic payment transaction information;

wherein the payer computing device interacts electronically with the server and the trusted consumer banking service computing device, wherein the payee computing device interacts electronically with the server, and wherein the server interacts electronically with one or more third party payment processing computer systems to record payments made by payers to payees, wherein recording payments updates a ledger representing one or more accounts held by a payer or a payee with the one or more third party payment processing computer systems, the computer-implemented transaction management method comprising:

receiving, by the server and from the payer computing device, an electronic payment transaction comprising a monetary value;

in response to identifying, by the server, the first financial account from which to debit the monetary value for the electronic payment transaction and based on receiving the electronic payment transaction, debiting the first financial account of the monetary value;

crediting, by the server, a holding account established at the server and associated solely with the payee for a credit amount, wherein the holding account is separate from the first financial account, wherein the credit amount is the monetary value for the electronic payment transaction;

generating, by the server, a unique hyperlink associated solely with the electronic payment transaction;

transmitting, by the server or the payer computing device, the unique hyperlink via an electronic communication to the payer computing device, wherein accessing the unique hyperlink on the payee computing device will cause the payee computing device to generate a payment acceptance interface uniquely associated with the electronic payment transaction for entry by the payee computing device of debit card information which associated with a second financial account associated with the payee and for crediting the monetary value to complete the electronic payment transaction, wherein the payment acceptance interface is generated on the payee computing device via a browser or a device-specific application already native to the payee computing device;

receiving, by the server and in response to the payee computing device accessing the unique hyperlink, a

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request to deliver the payment acceptance interface associated with the electronic payment transaction to the payee computing device;

in response to determining that the unique hyperlink is valid, delivering, by the server, the payment acceptance interface associated with the electronic payment transaction to the payee computing device;

receiving, by the server and based on payee interaction with the payment acceptance interface on the payee computing device, the debit card information associated with the second financial account associated with the payee computing device, wherein the payee interaction occurs without a requirement to create an account with a transaction management system, or log in to the transaction management system, or download any dedicated application associated with the electronic payment transaction;

based on receiving the debit card information, processing the electronic payment transaction by transmitting, by the server, a debit instruction to the holding account to debit the holding account for the monetary value, and a credit instruction to a second financial institution associated with the second financial account to post a credit to the second financial account for the monetary value; and

automatically, in response to processing the electronic payment transaction, transmitting, by the server and to the one or more third party payment processing computer systems via an electronic broadcast communication, payment transaction data between the payer and the payee.

12. The computer-implemented transaction management method of claim 11, wherein transmitting the unique hyperlink via the electronic communication to the payee computing device comprises providing, for display via a message graphical user interface for viewing the electronic communication, the unique hyperlink together with a text message element for generating a message to the payee computing device.

13. The computer-implemented transaction management method of claim 11, wherein receiving, by the server and from the payer computing device, the electronic payment transaction comprising the monetary value comprises:

providing, for display via a user interface of the application installed on the payer computing device, a payee selection element and a transaction value element; and

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identifying the monetary value and the payee computing device based on user interaction with the payee selection element and the transaction value element.

14. The computer-implemented transaction management method of claim 11, further comprising debiting the first financial account of the monetary value based on determining that the first financial account or one or more additional financial accounts corresponding to the payer computing device have sufficient funds to cover the electronic payment transaction.

15. The computer-implemented transaction management method of claim 14, further comprising determining, utilizing a machine learning model, a feasibility recommendation that the first financial account and the one or more additional financial accounts corresponding to the payer computing device have sufficient funds to cover the electronic payment transaction.

16. The computer-implemented transaction management method of claim 11, wherein delivering, by the server, the payment acceptance interface associated with the electronic payment transaction to the payee computing device comprises providing, for display:

a receive payment message indicating the monetary value and a payer identifier corresponding to the payer computing device, and

one or more debit card information entry fields.

17. The computer-implemented transaction management method of claim 11, wherein generating, by the server, the unique hyperlink associated with the electronic payment transaction comprises generating a hyperlink entry for the unique hyperlink within a hyperlink database.

18. The computer-implemented transaction management method of claim 17, wherein determining that the unique hyperlink is valid comprises, in response to determining that the payee computing device has accessed the unique hyperlink, accessing the hyperlink entry within the hyperlink database to verify that the unique hyperlink is valid.

19. The computer-implemented transaction management method of claim 11, further comprising:

determining an expiration trigger associated with the unique hyperlink; and

determining that the unique hyperlink is valid by determining that the expiration trigger has not occurred.

20. The computer-implemented transaction management method of claim 19, wherein the expiration trigger comprises a number of visits to the payment acceptance interface corresponding to the unique hyperlink.

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